

JSS 1-3

FARM-TO-FOOD ACTIVITY SYSTEM

FF-301 to FF-400

PRODUCTION -> QUALITY -> POST-HARVEST -> DATA -> LOSS/WASTE -> DECISION

Purpose	Connect school-garden learning to food origin, harvest, handling, quality, preparation pathways, food literacy and waste reduction.
Food-safety boundary	These are education activities, not authorization for unrestricted tasting or processing. School food-safety rules and teacher supervision take priority.
Core learning chain	GARDEN -> CROP -> HARVEST -> HANDLE -> SORT/MEASURE -> PREPARE/PROCESS PATHWAY -> FOOD -> CHOICE -> WASTE/RESOURCE
Connections	Use with GrowMeal practical cards, journals, mission cards, vocabulary and assessment items.

GARDEN 01 - Production Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-301 - FOOD ORIGIN

Garden	Production Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-302 - HARVEST READINESS

Garden	Production Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-303 - SAFE HANDLING

Garden	Production Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-304 - SORTING & QUALITY

Garden	Production Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-305 - FARM-TO-FOOD JOURNEY

Garden	Production Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-306 - MEASUREMENT & YIELD

Garden	Production Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-307 - STORAGE & LOSS

Garden	Production Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-308 - PREPARATION PATHWAY

Garden	Production Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-309 - FOOD CHOICE & LITERACY

Garden	Production Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-310 - WASTE-TO-RESOURCE

Garden	Production Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Production Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 02 - Raised Systems Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-311 - FOOD ORIGIN

Garden	Raised Systems Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-312 - HARVEST READINESS

Garden	Raised Systems Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-313 - SAFE HANDLING

Garden	Raised Systems Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-314 - SORTING & QUALITY

Garden	Raised Systems Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-315 - FARM-TO-FOOD JOURNEY

Garden	Raised Systems Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-316 - MEASUREMENT & YIELD

Garden	Raised Systems Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-317 - STORAGE & LOSS

Garden	Raised Systems Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-318 - PREPARATION PATHWAY

Garden	Raised Systems Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-319 - FOOD CHOICE & LITERACY

Garden	Raised Systems Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-320 - WASTE-TO-RESOURCE

Garden	Raised Systems Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Raised Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 03 - Grid Experiment Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-321 - FOOD ORIGIN

Garden	Grid Experiment Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-322 - HARVEST READINESS

Garden	Grid Experiment Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-323 - SAFE HANDLING

Garden	Grid Experiment Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-324 - SORTING & QUALITY

Garden	Grid Experiment Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-325 - FARM-TO-FOOD JOURNEY

Garden	Grid Experiment Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-326 - MEASUREMENT & YIELD

Garden	Grid Experiment Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-327 - STORAGE & LOSS

Garden	Grid Experiment Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-328 - PREPARATION PATHWAY

Garden	Grid Experiment Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-329 - FOOD CHOICE & LITERACY

Garden	Grid Experiment Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-330 - WASTE-TO-RESOURCE

Garden	Grid Experiment Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Grid Experiment Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 04 - Crop Rotation Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-331 - FOOD ORIGIN

Garden	Crop Rotation Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-332 - HARVEST READINESS

Garden	Crop Rotation Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-333 - SAFE HANDLING

Garden	Crop Rotation Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-334 - SORTING & QUALITY

Garden	Crop Rotation Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-335 - FARM-TO-FOOD JOURNEY

Garden	Crop Rotation Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-336 - MEASUREMENT & YIELD

Garden	Crop Rotation Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-337 - STORAGE & LOSS

Garden	Crop Rotation Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-338 - PREPARATION PATHWAY

Garden	Crop Rotation Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-339 - FOOD CHOICE & LITERACY

Garden	Crop Rotation Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-340 - WASTE-TO-RESOURCE

Garden	Crop Rotation Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Crop Rotation Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 05 - Intercropping Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-341 - FOOD ORIGIN

Garden	Intercropping Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-342 - HARVEST READINESS

Garden	Intercropping Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-343 - SAFE HANDLING

Garden	Intercropping Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-344 - SORTING & QUALITY

Garden	Intercropping Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-345 - FARM-TO-FOOD JOURNEY

Garden	Intercropping Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-346 - MEASUREMENT & YIELD

Garden	Intercropping Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-347 - STORAGE & LOSS

Garden	Intercropping Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-348 - PREPARATION PATHWAY

Garden	Intercropping Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-349 - FOOD CHOICE & LITERACY

Garden	Intercropping Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-350 - WASTE-TO-RESOURCE

Garden	Intercropping Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Intercropping Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 06 - Water-Smart Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-351 - FOOD ORIGIN

Garden	Water-Smart Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-352 - HARVEST READINESS

Garden	Water-Smart Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-353 - SAFE HANDLING

Garden	Water-Smart Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-354 - SORTING & QUALITY

Garden	Water-Smart Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-355 - FARM-TO-FOOD JOURNEY

Garden	Water-Smart Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-356 - MEASUREMENT & YIELD

Garden	Water-Smart Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-357 - STORAGE & LOSS

Garden	Water-Smart Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-358 - PREPARATION PATHWAY

Garden	Water-Smart Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-359 - FOOD CHOICE & LITERACY

Garden	Water-Smart Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-360 - WASTE-TO-RESOURCE

Garden	Water-Smart Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Water-Smart Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 07 - Compost & Soil Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-361 - FOOD ORIGIN

Garden	Compost & Soil Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-362 - HARVEST READINESS

Garden	Compost & Soil Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-363 - SAFE HANDLING

Garden	Compost & Soil Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-364 - SORTING & QUALITY

Garden	Compost & Soil Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-365 - FARM-TO-FOOD JOURNEY

Garden	Compost & Soil Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-366 - MEASUREMENT & YIELD

Garden	Compost & Soil Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-367 - STORAGE & LOSS

Garden	Compost & Soil Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-368 - PREPARATION PATHWAY

Garden	Compost & Soil Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-369 - FOOD CHOICE & LITERACY

Garden	Compost & Soil Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-370 - WASTE-TO-RESOURCE

Garden	Compost & Soil Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Compost & Soil Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 08 - Soilless Learning Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-371 - FOOD ORIGIN

Garden	Soilless Learning Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-372 - HARVEST READINESS

Garden	Soilless Learning Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-373 - SAFE HANDLING

Garden	Soilless Learning Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-374 - SORTING & QUALITY

Garden	Soilless Learning Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-375 - FARM-TO-FOOD JOURNEY

Garden	Soilless Learning Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-376 - MEASUREMENT & YIELD

Garden	Soilless Learning Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-377 - STORAGE & LOSS

Garden	Soilless Learning Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-378 - PREPARATION PATHWAY

Garden	Soilless Learning Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-379 - FOOD CHOICE & LITERACY

Garden	Soilless Learning Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-380 - WASTE-TO-RESOURCE

Garden	Soilless Learning Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Soilless Learning Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 09 - Sensor-Ready Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-381 - FOOD ORIGIN

Garden	Sensor-Ready Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-382 - HARVEST READINESS

Garden	Sensor-Ready Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-383 - SAFE HANDLING

Garden	Sensor-Ready Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-384 - SORTING & QUALITY

Garden	Sensor-Ready Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-385 - FARM-TO-FOOD JOURNEY

Garden	Sensor-Ready Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-386 - MEASUREMENT & YIELD

Garden	Sensor-Ready Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-387 - STORAGE & LOSS

Garden	Sensor-Ready Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-388 - PREPARATION PATHWAY

Garden	Sensor-Ready Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-389 - FOOD CHOICE & LITERACY

Garden	Sensor-Ready Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-390 - WASTE-TO-RESOURCE

Garden	Sensor-Ready Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Sensor-Ready Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

GARDEN 10 - Applied Systems Garden

Ten activities move learners from identifying where food comes from to reasoning about handling, quality, loss, choices and resource recovery.

FF-391 - FOOD ORIGIN

Garden	Applied Systems Garden
Learning objective	Analyze food origin as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-392 - HARVEST READINESS

Garden	Applied Systems Garden
Learning objective	Analyze harvest readiness as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-393 - SAFE HANDLING

Garden	Applied Systems Garden
Learning objective	Analyze safe handling as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-394 - SORTING & QUALITY

Garden	Applied Systems Garden
Learning objective	Analyze sorting & quality as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-395 - FARM-TO-FOOD JOURNEY

Garden	Applied Systems Garden
Learning objective	Analyze farm-to-food journey as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-396 - MEASUREMENT & YIELD

Garden	Applied Systems Garden
Learning objective	Analyze measurement & yield as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-397 - STORAGE & LOSS

Garden	Applied Systems Garden
Learning objective	Analyze storage & loss as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-398 - PREPARATION PATHWAY

Garden	Applied Systems Garden
Learning objective	Analyze preparation pathway as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-399 - FOOD CHOICE & LITERACY

Garden	Applied Systems Garden
Learning objective	Analyze food choice & literacy as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?

FF-400 - WASTE-TO-RESOURCE

Garden	Applied Systems Garden
Learning objective	Analyze waste-to-resource as a production and post-harvest problem in the Applied Systems Garden, quantify relevant evidence and justify a management decision.
Materials	Investigation journal, sample IDs, balance/measurement tools where available, clean containers, quality checklist, calculator.
DIY / low-cost	Use locally available baskets, clean reused food-grade containers, handmade labels, paper process cards and simple balances/measuring vessels where suitable.
Activity pathway	1. Define the farm-to-food problem. 2. Establish sample and quality criteria. 3. Collect traceable data. 4. Quantify yield, quality class, loss or process performance as appropriate. 5. Identify a critical loss/handling point. 6. Recommend a management response.
Evidence to record	Traceable dataset, quality/loss calculation, process map and qualified recommendation.
Farm-to-Food challenge	Define a measurable threshold that would trigger the recommended action.
Safety	No tasting unless the school has separately approved a supervised food activity. Wash hands before and after handling produce. Use only clean, teacher-approved samples and containers. Teachers control knives, heat, electrical equipment, chemicals and any processing machinery.
Journal connection	Record the evidence, sequence, measurement, process map or reflection in the level-appropriate GrowMeal journal.
Mission/assessment	Use the evidence to complete a related Garden Mission and assess knowledge, practical, data or problem-solving competence.

Reflection: Where could food quality be protected, waste reduced, or the garden-to-food connection explained more clearly?